Evaluating Self-Selection Policing: An Agent-Based Modelling Approach to Patrol Strategy Performance

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UNIVERSITY COLLEGE LONDON

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BSc Dissertation

STUDENT'S DECLARATION

I, Roy Wang, hereby declare that this dissertation is my own original work and that I have clearly identified and acknowledged all source material used. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other examination.

This dissertation is 9847 words in length (including abstract but excluding reference list and reasonable use of tables and figures).

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Date: 28/04/2025

University College London
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SUPERVISOR'S DECLARATION

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I confirm that the student named above has undertaken this dissertation under my supervision, attended meetings with me as requested and provided me with adequate information about the progress of the research.

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Title of proposed project: Can Self-Selection be Applicable to Catch Criminals? An Agent-Based Simulation Experiment

Project ID: 943

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Description of use of AI:

During the Literature Review stage, I used ChatGPT to provide feedback on the paragraph structure, which I then adapted and incorporated into my dissertation. Specifically, ChatGPT assisted in suggesting a logical flow: beginning with an overview of traditional policing strategies and their limitations, introducing the theoretical foundation of modelling Self-Selection Policing, and subsequently linking the discussion to the application of ABM as a tool for evaluating policing strategies. I also used ChatGPT for grammar checking throughout the dissertation and for testing my code during the development and testing of the agent-based model.

Abstract

An agent-based model (ABM) was developed to evaluate the effectiveness of Self-Selection Policing (SSP) compared to Hot Spots Policing (HSP) and Random Patrol strategies. The model simulated offender and police interactions across 500 steps over 50 replications per strategy. SSP demonstrated higher detection precision and lower recall rates relative to the alternatives. Sensitivity analysis showed that intensified SSP operations improved detection rates while reducing reliance on selective targeting of high-risk offenders. The key findings of this study include: (1) experimental simulation evidence on the relative strengths and weaknesses of SSP compared to HSP and Random strategies; (2) the precision–recall trade-off inherent in SSP; and (3) how investigative intensity and bias alter the sources of successful detections under SSP. The main implication is (4) supporting the use of agent-based modelling for mechanism-level evaluation of policing strategies beyond aggregate outcome measures.

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1. Introduction

Accurately identifying serious and persistent offenders presents a fundamental challenge for policing. Traditional investigation methods, notably the suspect-centred approach, focus predominantly on individuals already known to law enforcement (Maguire, 2008; McConville et al., 1993). While this strategy can be effective in certain circumstances, it has significant limitations. These include a potential prioritisation of securing convictions over protecting individual rights (Maguire, 2008), and a tendency to neglect alternative investigative leads (Tong, Bryant, and Horvath, 2009). Furthermore, the classification of prolific offenders is often imprecise. Research indicates that some individuals are incorrectly labelled, while genuinely active offenders are overlooked, and patterns of offending may fluctuate over time (Townsley and Pease, 2002).

Self-Selection Policing (SSP) proposes a different investigative logic. Instead of centring inquiries on known individuals or specific locations, SSP focuses on observable minor infractions that may indicate broader criminal activity (Roach and Pease, 2014; 2016). The approach is grounded in the proposition that serious offenders tend to display behavioural versatility rather than specialisation, committing a range of offences across different categories. This assertion is supported by criminological theories and empirical findings. For instance, longitudinal studies of offending careers have demonstrated that individuals with early-onset offending often engage in diverse criminal behaviours throughout their lives (Blumstein et al., 1986; Piquero et al., 1999). Analysis of offence sequences further reveals that individuals convicted of serious crimes frequently transition into different crime types rather than remaining confined to a single category (Tarling, 1993).

Theoretical perspectives from environmental criminology further support the plausibility of the SSP framework. Rational Choice Perspective, Routine Activity Theory, and Crime Pattern Theory collectively suggest that offenders make decisions based on situational opportunities, daily routines, and familiarity with specific activity spaces (Cohen and Felson, 1979; Brantingham and Brantingham, 1993; Cornish and Clarke, 2008). These theories imply that offenders, when exposed to conducive environments, are likely to engage in a variety of offences as opportunities present themselves. Within this context, minor offences may offer observable entry points for identifying individuals involved in more serious criminal activities (Farquharson, 2020; Roach and Pease, 2016).

Although the theoretical foundation and initial observational evidence for SSP are compelling, significant questions remain regarding its operational effectiveness. Most existing evaluations are small in scale and observational in nature, limiting the ability to establish causal relationships (Chenery, Henshaw, and Pease, 1999; Roach and Hatton, in press). Moreover, practical concerns, such as the risk of false positives—where individuals committing minor offences without broader criminal involvement are mistakenly targeted—have been raised (Rossmo, 2020). Official crime data, often shaped by underreporting and hierarchical crime counting rules, further obscure the relationships that SSP seeks to exploit (Skogan, 1977; Buil-Gil et al., 2021). As a result, an evaluation of SSP's mechanisms is necessary to determine under what conditions the approach can effectively contribute to serious offender detection. This includes examining the role of investigative intensity. To explore these concerns in a controlled and mechanism-focused way, this study poses the following research question:

Research Question (RQ):

Can Self-Selection Policing improve serious offender detection while balancing investigative effectiveness and resource efficiency?

To investigate this question, the current study employs agent-based modelling (ABM). ABM provides a computational framework for simulating the dynamic interactions between offenders and police agents within structured but evolving environments (Eck and Liu, 2008; Groff and Birks, 2008). This approach allows for the modelling of heterogeneous agent behaviours (i.e., individuals with differing traits or decision rules) over time, offering the opportunity to test how minor infractions, risk perceptions, and policing strategies interact to influence detection outcomes. Through simulation, the study isolates the mechanisms underpinning SSP and systematically compares its performance with other investigative strategies under varied conditions.

2. Literature Review

2.1 Traditional and Emerging Police Strategies

Police investigations have traditionally focused on individuals already known to law enforcement, with investigations often built around these suspects (Maguire, 2008). These individuals usually have a long history of offending and are expected to either confess or provide information when questioned. This method is known as the "suspect-centred" approach (McConville et al., 1993). While this approach can be effective in some cases, it has several limitations. For instance, Maguire (2008) argues that the strategy may prioritise securing convictions over protecting individual rights, particularly when dealing with those already presumed guilty. Similarly, Tong, Bryant, and Horvath (2009) note that exclusive reliance on this method may result in missed investigative opportunities because alternative leads are often left unexplored.

Townsley and Pease (2002) also raise concerns regarding how police identify prolific offenders. Misclassifications can occur; individuals may be wrongly labelled as prolific, while genuinely active offenders might be overlooked. Even among those accurately identified, offending patterns may vary over time. A person who is highly active during one period might later reduce their level of offending. Moreover, given the frequency of co-offending, imprisoning one individual does not necessarily prevent their associates from continuing criminal activity.

Humphrey and Gibbs Van Brunschot (2017) add that both the justice system and the public often assume that certain groups of offenders specialise in particular types of crime, although evidence suggests otherwise. Such assumptions can render specialised police units less effective if they are based on inaccurate beliefs about offender profiles.

In contrast, focusing police efforts on areas with high crime rates, commonly referred to as hot spots policing, can improve the likelihood of detecting and apprehending offenders in those locations. This strategy may also increase offenders' perception of risk, thereby influencing their decision-making (Nagin, Solow, and Lum, 2015). When individuals believe the chance of being caught is higher, they may be deterred from acting on criminal opportunities because the perceived risk outweighs the potential gain (Nagin, Solow, and Lum, 2015).

However, studies examining the spatial distribution of different crime types have questioned the practical effectiveness of hot spots policing. Haberman (2017), for example, found that even when using multiple methods to identify hot spots, the Philadelphia Police Department would need to monitor between 1,001 intersections (covering nearly 12 square miles) and 3,160 aggregated clusters (just under 6 square miles) to target all hot spots across 11 crime types. This suggests that the areas requiring police attention can be extensive. If similar spatial patterns are found elsewhere, it implies that implementing hot spots policing for a wide range of crime types would still demand substantial resources (Rosenbaum, 2006; Brage et al., 2012; Haberman, 2017).

Another strategy, referred to as "sting" operations, targets individuals believed to be frequent offenders by monitoring specific locations or behaviours (Chenery, Henshaw, and Pease, 1999). However, these tactics depend heavily on the availability of accurate knowledge regarding crime patterns, which is not always assured. In some cases, this type of surveillance can also be perceived as harassment. The authors argue that focusing on offenders' actions, rather than their identity or location, may offer a more reliable approach. This line of reasoning closely aligns with the concept underpinning Self-Selection Policing (SSP).

Roach and Pease (2014) examined how police officers assess whether individuals specialise in particular types of crime. Their study found that many officers wrongly assume that a person's past offences reliably predict future behaviour within the same category. This overestimation can be dangerous. For example, individuals committing minor offences might, in fact, be involved in more serious crimes but could be overlooked because they do not match the expected profile. Roach and Pease (2014) further note that this misjudgement could place officers at risk if they underestimate a suspect based on a minor offence and fail to take appropriate precautions. Recognising that many offenders are versatile, rather than specialised, may help improve the effectiveness of police investigations.

2.2 Offender Versatility

Roach and Pease (2016) identify several key assumptions that underpin the use of SSP as an investigative method. A central idea is that serious and active offenders are typically not offence-specific but instead display versatility in their criminal behaviour. Rather than specialising in a single type of crime, these individuals tend to commit a range of offences, including both minor and serious ones. This behavioural pattern is critical to the logic of SSP because it allows relatively low-level infractions to serve as entry points for identifying those engaged in more serious criminal activity. Roach and Pease (2016) further develop the concept of offender versatility by distinguishing between horizontal and vertical versatility.

Horizontal versatility refers to the tendency of offenders to engage in different types of crime across various categories (Roach and Pease, 2016). Drawing on the general theory of crime, Gottfredson and Hirschi argue that individuals with low self-control are unlikely to specialise in one type of offending and instead tend to engage in a variety of criminal behaviours depending on the situation and the opportunities available (Piquero et al., 1999). This theory suggests that the same individual may commit different types of crime over time, such as robbery, car theft, and burglary, which reflects a lack of offence specialisation.

The relationship between low self-control and offence variety has been considered strong enough that the number of offence types committed can serve as a proxy measure of an individual's level of self-control (Hirschi and Gottfredson, 1995). Earlier research also shows that those who begin offending at a younger age are more likely to have less specialised criminal careers (Blumstein et al., 1986; Piquero et al., 1999). Collectively, these findings support the idea that serious offenders are often not confined to a single crime category but instead display versatility across offence types. This provides theoretical justification for using minor infractions as potential entry points to detect more serious offending behaviour, which is a key mechanism underpinning the SSP approach (Roach and Pease, 2016). Furthermore, Farrington et al. (2013) identify offences such as benefit fraud, obstructing police, and shop theft as examples of minor crimes that may serve as useful indicators for identifying versatile offenders.

Evidence for horizontal versatility, meaning offenders engaging in multiple types of crime rather than specialising, can be observed in offence sequence data. Tarling (1993), for example, examined the likelihood of offenders transitioning from one crime category to another following a Crown Court index offence as shown in Table 1.

Table 1. Probability of Sequences of Offence Types in those with a Crown Court Index Offence (Roach and Pease, 2016)

Offence type	Violence%	Sex%	Burglary%	Theft%	Other%
Violence	26	2	19	30	24
Sex	7	25	19	38	12
Burglary	8	1	13	34	14
Theft	10	2	27	45	16
Other	14	1	21	35	29

Modified from Tarling (1993)
Rows represent kth offence type, columns k + 1th offence type

The data suggest that criminal behaviour often spans multiple offence types rather than remaining within a single category. For instance, among individuals initially convicted of violent crimes, only 26% later committed another violent offence, whereas a substantial proportion transitioned into theft or burglary. A similar pattern was observed among sex offenders, with many subsequently engaging in non-sexual offences such as theft. These patterns reinforce the foundational logic behind SSP, namely that individuals involved in serious crimes also participate in minor offending, providing practical opportunities for early detection (Roach and Pease, 2016).

Vertical versatility refers to the idea that offenders do not necessarily abandon minor offences as they progress in their criminal careers. This challenges the common assumption of linear escalation, often described as the "graduation hypothesis", which suggests that individuals move from minor to more serious crimes over time (Blumstein et al., 1986). If this model held true in a strict sense—meaning that serious offenders no longer engaged in low-level offending—then the logic behind SSP would be undermined. However, research suggests that escalation does not necessarily imply exclusivity. Roach and Pease (2016) argue that as long as minor crimes remain part of the overall offence pattern, even alongside more serious ones, the premise of SSP remains valid.

While the notion of escalation is widely accepted as a general trend in criminal development (Blumstein et al., 1986), scholars such as Loeber and Le Blanc (1990) have proposed a more nuanced understanding. They point out that changes in criminal behaviour occur in both quantitative and qualitative ways, including fluctuations in offence type, frequency, and severity. Escalation is not always linear or permanent. Offenders may scale back their activity due to incarceration, changes in opportunity, or personal choice. In some cases, they may alternate between more and less serious offences throughout their criminal careers. Mazerolle et al. (2000) observed that individuals who start offending at an early age and persist into adulthood tend to exhibit a broader range of offending behaviours compared to those who cease offending earlier.

This broader perspective supports the idea that serious offenders continue to engage in minor infractions, rather than exclusively committing high-level crimes. Le Blanc and Fréchette (1989) offer a definition of escalation that captures this variation, describing it as a shift through different forms of delinquency rather than a one-directional climb in seriousness (cited in Le Blanc, 2002). Framing escalation in this way reinforces the relevance of SSP. If serious offenders maintain involvement in lower-level offences, these actions can still serve as useful triggers for police intervention. Roach and Pease (2016) further argue that offending patterns are better understood as a distribution, where high-severity offences often coexist with a broader set of minor crimes.

2.3 Early Empirical Support for Self-Selection Policing (SSP)

Early empirical support for the logic of SSP comes from research into the use of minor traffic offences as potential indicators of more serious criminal behaviour. A study examining vehicles illegally parked in disabled bays found that 21% attracted immediate police attention, compared to only 2% of those parked legally (Chenery, Henshaw, and Pease, 1999). Additionally, 18 percent of the illegally parked vehicles had previously been involved in criminal activity, whereas none of the legally parked ones had such links. In total, 25 percent of the illegally parked vehicles warranted police action, compared to only 2 percent of the legally parked vehicles. Following these findings, this self-selection approach was incorporated into police intelligence work in Huddersfield, where further checks revealed that approximately one-third of illegally parked vehicles were linked to various offences, including drug possession, assault, vehicle crime, theft, and burglary. These results offer preliminary support for the use of SSP alongside conventional policing strategies.

Further evidence is provided by a pilot study conducted by Roach and Hatton (in press), reviewed by Farquharson (2020), which examined the potential of traffic-related behaviours such as dangerous driving and parking violations as self-selection triggers. In this study, officers stopped vehicles based on observed driving behaviour, resulting in a sample of 36 individuals, of whom 20 had previous convictions. Although the findings lend some support to the principle that minor infractions may indicate involvement in more serious offences, the absence of a control group limits the conclusions that can be drawn about the relative offending rate of the sample. Nonetheless, the detection of vehicles associated with Organised Crime Groups, including two with firearms markers, and the arrest of five individuals suggests that the approach has operational value, even when applied at a small scale (Farquharson, 2020). These outcomes also suggest the presence of vertical versatility in offending, as some individuals identified through minor traffic violations were also involved in more serious organised crime.

A recognised challenge in the practical application of SSP is the issue of false positives (Rossmo, 2020). Although the approach is designed to identify trigger offences that signal active or serious offenders, many individuals who are not of particular interest to police may commit the same low-level infractions. This is partly due to the large number of people in the general population who engage in minor legal violations without any connection to more serious criminal behaviour. As Rossmo (2020) notes, “even the best trigger offence will identify more people who are not serious offenders than people who are so” (p. 132). Enhancing the sensitivity of SSP—meaning its ability to increase true positive identifications—remains an important area for future research and refinement. In this regard, findings from environmental criminology, which consistently highlight the spatial and temporal patterns of offending, may offer valuable insights for improving SSP practices (Rossmo, 2020).

2.4 Using Agent-Based Modelling (ABM) to Simulate SSP

Early work by Brantingham and Brantingham (2004) helped to establish the relevance of ABM within criminological theory, particularly in relation to opportunity-based explanations of crime. Building on this foundation, Groff, Johnson, and Thornton (2019) identified more than forty published ABMs focused on urban crime, reflecting a growing methodological interest in simulation-based approaches. Several researchers have since highlighted ABM's value as an analytical tool for testing crime-reduction strategies in contexts where controlled field experiments may be infeasible or ethically constrained (Groff and Mazerolle, 2008; Groff and Birks, 2008; Johnson, 2009; Verma et al., 2013).

Brantingham and Brantingham's (2004) influential work played a key role in stimulating interest in the use of ABM to explore opportunity theories of crime. In a subsequent review, Groff, Johnson, and Thornton (2019) identified 43 ABMs focused on urban crime, reflecting the expanding application of simulation techniques within criminology and crime science. Interest in ABM as a methodological approach has continued to grow, with several scholars emphasising its value for assessing crime-reduction strategies in contexts where conventional experimental designs are impractical (Groff and Mazerolle, 2008; Groff and Birks, 2008; Johnson, 2009; Verma, Ramyaa, and Marru, 2013).

One of the most critical advantages of ABM is its ability to generate counterfactuals for large-area impacts and to allow for dynamic modelling over time. In contrast to traditional experiments, ABM allows the same agent-based environment to be simulated under multiple policy scenarios, enabling the generation of true counterfactuals even in the presence of complex interdependencies and feedback mechanisms (Marshall and Galea, 2015). This provides additional rigour by enabling the exploration of potential outcomes across multiple trials.

Given this methodological potential, the present study applies ABM to develop a more dynamic understanding of the impacts of Self-Selection Policing (SSP). It constructs an agent-based simulation of offenders committing both serious and minor offences and tests the effects of different SSP interventions on overall crime rates, serious crime rates, and true positive detections under varied assumptions (Eck and Liu, 2008; Groff and Birks, 2008). This approach enables the design of artificial yet realistic environments in which heterogeneous agents make decisions in response to structured but evolving conditions.

To ensure that simulated offender behaviour within the ABM is theoretically grounded, it is essential to incorporate insights from environmental criminology. While individual traits are often highlighted as factors influencing criminal behaviour, environmental criminology emphasises the role of external, situational, and contextual factors in shaping offending patterns (Hollin, 2006; Quinsey, 1995). Within this framework, three key opportunity theories are particularly relevant to the SSP model: Rational Choice Perspective (RCP), Routine Activity Theory (RAT), and Crime Pattern Theory (CPT) (Cohen and Felson, 1979; Brantingham and Brantingham, 1984; Cornish and Clarke, 2008). These perspectives offer structured frameworks for modelling how offenders perceive risk, identify opportunity, and move through space. They emphasise decision-making processes, deterrence mechanisms, daily routines, and spatial familiarity, which together provide a theoretical foundation for constructing credible behavioural rules in agent-based simulations of self-selection policing.

2.5 Environmental Criminology Theories Relevant to SSP

2.5.1 Rational Choice Perspective

Rational Choice Perspective (RCP) emphasises the role of situational and environmental factors in shaping offender decision-making (Cornish and Clarke, 2008). It assumes that individuals engage in crime after weighing the perceived benefits against the risks associated with a particular act. Within the SSP framework, this assumption implies that those who have previously justified committing a serious offence are unlikely to be deterred from engaging in minor offences, as the perceived risks associated with such behaviour are typically lower. RCP further supports the concept of offender versatility by proposing that it is the availability of opportunity, rather than the specific offence type, that drives criminal behaviour. Offenders are therefore expected to act on opportunities as they arise, rather than selectively pursuing crimes they have committed previously (Roach and Pease, 2016).

Nonetheless, critiques of RCP have pointed out its limitations in explaining certain non-instrumental or affect-driven offences, such as spontaneous violence or expressive acts (Pratt et al., 2006). Other scholars have also observed that the theory tends to underplay psychological factors such as impulsivity, moral reasoning, and shame (DeHaan and Vos, 2003). However, these critiques are more relevant to the internal motivations underlying specific acts and do not necessarily undermine the applicability of RCP to SSP. Since SSP focuses on the observable relationship between offending opportunity and behavioural versatility, its reliance on RCP remains theoretically appropriate. The theory's emphasis on how situational factors shape opportunistic offending across different crime types provides a critical foundation for the strategic identification of serious offenders through minor infractions.

2.5.2 Routine Activity Theory

According to Routine Activity Theory (RAT), criminal acts occur when three factors align: the presence of a motivated offender, the availability of a suitable target, and the lack of a capable guardian (Cohen and Felson, 1979). This framework highlights the situational nature of crime and emphasises that the likelihood of offending is shaped less by individual motivation alone and more by the environmental conditions that enable or prevent criminal activity. Daily routines, such as commuting patterns or social interactions in public spaces, may create opportunities for crime by increasing separation from guardianship or by exposing individuals to unfamiliar contexts (Cohen and Felson, 1979).

The model has been expanded through Eck's crime triangle (Felson and Eckert, 2018), which formalises the interaction between the offender, the crime target, and the place where the crime occurs. Surrounding this inner triangle is a protective triangle composed of three supervisory roles: the handler, who manages offenders; the place manager, who controls the setting; and the guardian, who protects the target. This development further illustrates how situational factors influence the occurrence of criminal events.

RAT provides useful insights for SSP. If crimes are shaped by opportunities arising in everyday environments, then serious offenders, when exposed to suitable targets and lacking oversight, are just as likely to engage in minor infractions as they are in serious ones (Farquharson, 2020). These minor offences, often low-risk and routine in nature, occur more frequently under such converging conditions. As a result, minor crimes may serve as practical indicators of more serious offending, supporting the logic behind using them as self-selection triggers. Pease (2006) introduces the idea of affordance, referring to the perception of opportunities for action in specific settings. Individuals with a criminal mindset are more likely to recognise these opportunities, particularly when environmental controls are weak. From the standpoint of SSP, this suggests that repeat or serious offenders are more attuned to perceiving and acting on a broad spectrum of offence opportunities present in daily life. Thus, when the conditions described by RAT are present, regardless of crime type, such individuals are likely to offend. This perspective helps explain both the versatility of their offending and the effectiveness of minor infractions in identifying them.

2.5.3 Crime Pattern Theory

Crime Pattern Theory (CPT) offers an environmental perspective on where crimes are likely to occur, focusing on how offenders navigate space in ways that shape their offending patterns (Brantingham and Brantingham, 1994). According to the authors, a crime event will only occur in an area where an individual is 'comfortable or sure of what will happen', with offenders relying on a mental map of locations they consider suitable for committing offences (Brantingham and Brantingham, 1993, p. 273). These locations, referred to as "activity spaces", are formed through routine movement between primary nodes such as home, workplace, and leisure or shopping areas (Brantingham and Brantingham, 1993).

This spatial logic aligns closely with the foundations of SSP. If criminal events are largely shaped by familiarity and accessibility, then individuals committing serious crimes within or between these known nodes may also engage in minor offences as part of their everyday routines. Roach and Pease (2016) suggest that such routine minor infractions not only occur more frequently but may also serve as indicators of more serious offending. In this context, minor crimes committed within a person's activity space may be especially useful to police as potential self-selection triggers.

CPT also intersects with concepts from Rational Choice Perspective (Cornish and Clarke, 2008), as Brantingham and Brantingham (1993) describe offender decision-making as "normal/rational in the context of committing a crime". This reinforces the idea that offending decisions are shaped by perceived opportunity and environmental familiarity. Furthermore, Pease's (2006) concept of "affordance", which refers to the recognition of a setting as offering a criminal opportunity, adds further depth to this view. Offenders, particularly those with greater criminal experience, may be more likely to identify specific locations or moments within their activity spaces as chances to offend. These opportunities are often associated with minor offences, but the same perceptual processes may also lead to serious crimes under favourable situational conditions (Farquharson, 2020; Roach and Pease, 2016). In this way, CPT provides both a spatial and behavioural rationale for understanding how serious and minor offending can co-occur in everyday environments, thereby lending further theoretical support to the SSP framework.

3. Methods

3.1 ABM Experiment Setup

The ABM was executed under three experimental patrol conditions: Self-Selection Policing (SSP), Hot Spots Policing (HSP), and a Random Patrol baseline. Each condition was designed to reflect a plausible deployment strategy observed in urban policing contexts. To assess the relative effectiveness of these strategies, all conditions were run under identical structural parameters, with the only difference being the patrol logic assigned to police agents.

For each patrol strategy, 50 simulation runs were carried out, with each run extending over 500 steps. This duration was considered sufficient for analysis, as the model's outputs show little variation between $t = 500$ and $t = 1000$ (Ten et al., 2016). A fixed random seed was used to ensure reproducibility across replications. The simulated grid environment follows Weisburd et al. (2017), measuring 100 × 100 grid cells per police beat, and was set with 100 criminal agents—consistent with 40% of the articles reviewed by the author—and 20 police agents, corresponding to 67% of the articles reviewed (Groff et al., 2019).

3.2. Criminal Agents

Criminal agents in the model make offence decisions at each simulation step through a bounded rationality framework. As shown in Figure 1, agents move to a neighbouring grid cell and assess local policing presence and environmental risk. If no deterrent factors are detected, agents calculate offence-specific scores, balancing perceived rewards and policing risks based on rational choice perspectives (Cornish and Clarke, 1986). An offence is initiated only if the Threshold Score (T) surpasses calculated thresholds, with additional stochastic elements introduced to reflect behavioural heterogeneity (Weisburd et al., 2017).

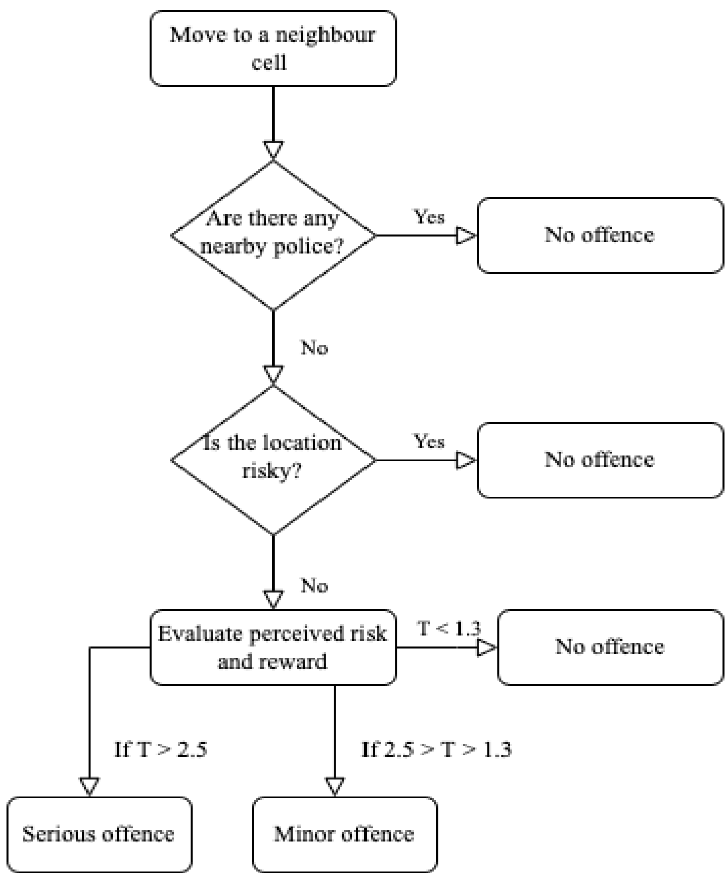

Figure 1. Decision-Making Process for Offence Initiation by Criminal Agents in ABM

This offence score integrates three key components: the type of offence (reflected in the associated reward), the environmental opportunity of the location, and the agent's individual perception of policing risk (Weisburd et al., 2017). A criminal action is initiated only when this score exceeds a behavioural threshold. This logic is first illustrated below and then expressed more formally.

$$Threshold = \underbrace{Reward}_{\text{offence type}} \cdot \underbrace{Opportunity(x, y, t)}_{\text{grid cell}} \cdot \underbrace{Propensity \cdot PercievedRisk}_{\text{individual} \cdot \text{environment}} \cdot \alpha$$

Formally, the offence threshold for offence type $i \in$ {serious, minor} is defined as:

$$T_i = R_i \cdot O(x, y, t) - \lambda \cdot PR \cdot \alpha_i$$

R_i is the reward associated with the offence type (3.0 for serious offences and 1.0 for minor offences), while O (x, y, t) represents the opportunity factor of the grid cell at location (x, y) and time t, which ranges from 0.5 to 1.5. The product $\lambda \cdot PR \cdot \alpha_i$ represents the total perceived penalty, where λ is the agent-specific propensity parameter indicating the initial motivation to commit crimes, uniformly sampled from (0.5, 1.5) at model initialisation; PR is the dynamically updated perceived policing risk; and α_i is the punishment multiplier (3.0 for serious and 0.5 for minor offences, respectively). Full details of the decision framework and equations are provided in Appendix 1.

3.3 Patrol Strategies

In the Self-Selection Policing (SSP) condition, police agents patrolled the grid with a predefined investigative bias, prioritising the identification of offenders associated with either minor or serious offences. Investigations were probabilistic and conditioned on recent behavioural records stored within a limited memory window. Successful investigations triggered a review of the offender's past activity, and individuals with multiple serious offences were classified as serious crime detections. Operational costs were assigned based on offence severity, and all investigative actions were systematically recorded for evaluation. Full implementation details, including decision rules and parameter specifications, are provided in Appendices 2 and 3.

In the Random Patrol condition, police agents were assigned to randomly selected grid cells within their vision radius at each simulation step. Patrol movements were fully random and uninformed by any prior incidents.

In the Hot Spots Policing (HSP) condition, a set of high-opportunity grid cells (*hotspot_cells*) was predefined based on static opportunity values assessed at model initialization. Officers assigned to this condition followed a fixed, looped movement pattern to provide continuous patrol coverage across these hotspot locations. Throughout the simulation, hotspot locations were dynamically updated in response to crime patterns that emerged during the simulation runs (Weisburd et al., 2017).

3.4 Model Design and Implementation

Building on existing theoretical frameworks and empirical studies of Self-Selection Policing (e.g., Roach and Pease, 2016; Farquharson, 2020; Rossmo, 2020), this study developed an original ABM to operationalise and evaluate the strategic logic of SSP under controlled simulation conditions. The model design, including agent decision processes, detection mechanisms, and intervention dynamics, was independently conceptualised by the author based on a synthesis of the literature and an interpretation of SSP's underlying behavioural assumptions. To the best of my knowledge, no prior ABM has been developed to systematically simulate SSP mechanisms in this manner. Detailed modelling procedures are described in Appendix 3.

3.5 Data Collection and Analysis

Across all patrol strategies, model performance was tracked through the systematic collection of key outcome metrics at each simulation step. These metrics included detection counts, precision, operational costs, and offender outcomes. All recorded values were averaged across replications within each condition. Descriptive statistics, inferential testing, and linear regression analyses were employed to evaluate the comparative effectiveness of the strategies. Outcome metrics were reported using means, standard deviations, and confidence intervals, with full model output provided in Appendix 5.

To assess statistical significance across strategies, one-way ANOVA tests were conducted for the main dependent variables, as described in the following section. Linear regression models incorporating interaction terms were applied to examine cost-efficiency trends and the accumulation of detections over simulation time. Paired-sample t-tests were used for direct comparisons between SSP variants and benchmark strategies under controlled random seeds, and corresponding confidence intervals were reported.

Sensitivity analyses were conducted to explore how variations in investigative initiative, detection accuracy, targeting bias, and memory depth affected the attribution of serious offender detections to the internal SSP mechanisms. Comparisons between the SSP_Low and SSP_High configurations are detailed in Appendix 4. Full Python code for experiment setup, execution, and result generation is available in Appendix 6.

4. Results

4.1 Detection Metrics Comparison of Patrol Strategies

Four core detection metrics were used to compare performance differences among the three patrol strategies. As shown in Figure 2, the RANDOM strategy achieved the highest mean number of detected serious offenders (M = 39.20), followed closely by SSP (M = 38.40) and HSP (M = 23.74), indicating a higher volume of serious offender identifications under RANDOM and SSP relative to HSP. However, SSP demonstrated the highest detection precision (M = 0.56), outperforming RANDOM (M = 0.41) and HSP (M = 0.30). **This indicates that, among detected offenders, SSP was more selective, capturing a higher proportion of serious offenders relative to total investigations, with a detection precision approximately 36.6% higher than that achieved by RANDOM.**

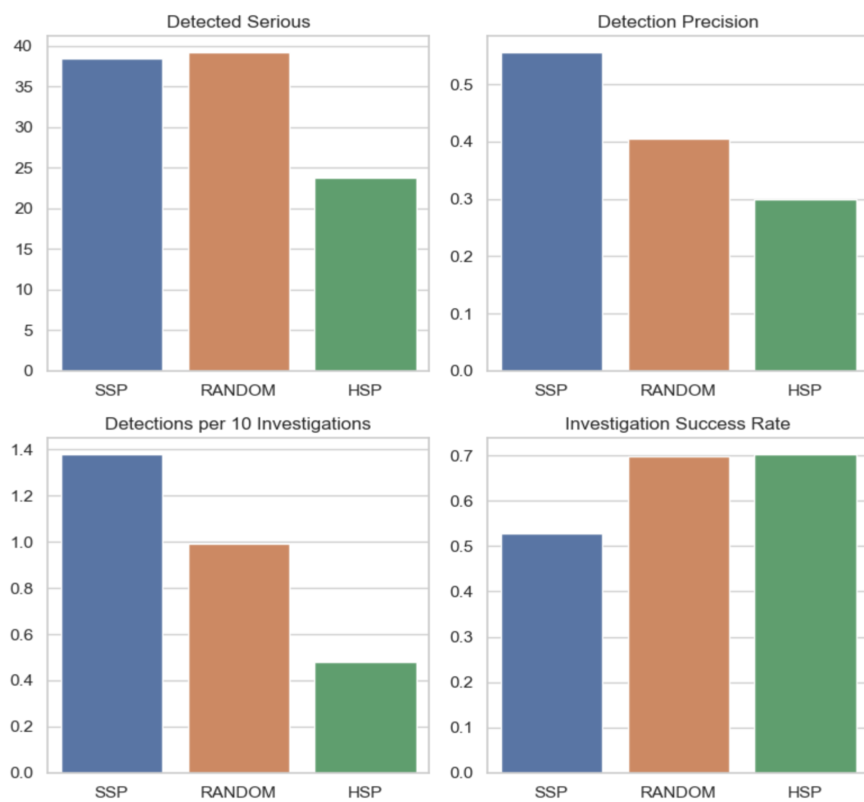

Figure 2. Comparative detection metrics across SSP, RANDOM, and HSP, including serious offender detections, detection precision, investigative yield, and investigation success rates.

When results were normalised per ten investigative actions, SSP again recorded the highest serious offender detections per 10 investigations (M = 1.38), compared to RANDOM (M = 0.99) and HSP (M = 0.48), reflecting greater investigative yield. Nevertheless, HSP achieved the highest investigation success rate (M = 0.703), narrowly surpassing RANDOM (M = 0.698), while SSP lagged behind at M = 0.53. This pattern likely reflects the different targeting logics: SSP prioritises suspects based on risk filtering rather than immediate apprehension probability, leading to a lower per-investigation success rate but a higher cumulative detection efficiency over time.

Taken together, these results highlight distinct operational trade-offs. SSP achieved superior precision and a higher yield per investigative effort, whereas RANDOM and HSP strategies achieved broader detection coverage and higher immediate success rates. No single strategy consistently dominated across all evaluated metrics.

4.2 Cost Efficiency and Cumulative Detection Over Time

To evaluate the cost-efficiency of different patrol strategies, Figure 3 presents the temporal evolution of average detection cost across 500 simulation steps. SSP persistently achieved the lowest average cost per detection throughout the simulation, reflecting greater economic efficiency relative to RANDOM and HSP. In contrast, HSP incurred the highest detection costs, with substantial variability across replications, while RANDOM maintained an intermediate profile characterised by moderate cost levels and a relatively stable downward trend.

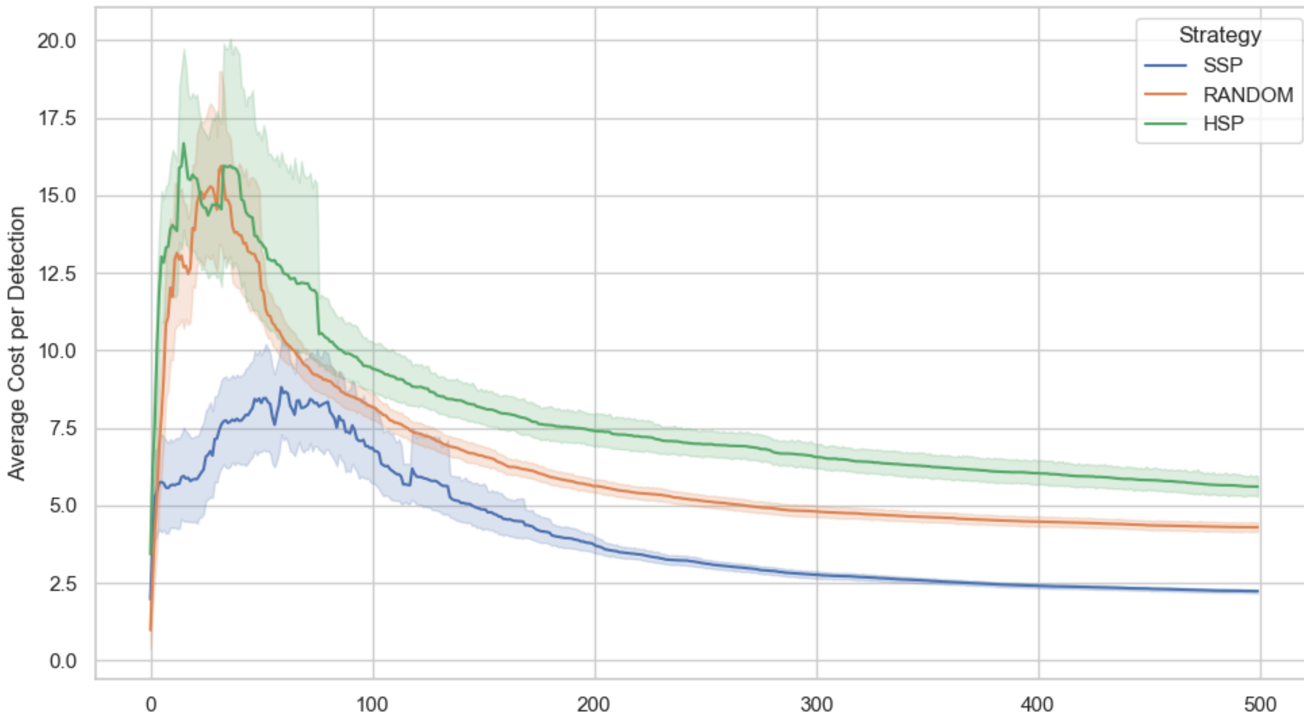

Figure 3. Average cost per detection across police patrol strategies over 500 simulation steps, showing mean trends and variability bands.

To formally assess these trends, a linear regression model with interaction terms was fitted. The regression results confirm that SSP consistently outperforms both RANDOM and HSP in cost efficiency over time. While all strategies show declining detection costs, SSP starts from a lower baseline ($\beta = 7.04$, $p < .001$) and demonstrates a steady downward trend ($\beta = -0.01195$, $p < .001$). In contrast, RANDOM and HSP both begin with significantly higher baseline costs—$\beta = 3.16$ and 4.90, respectively—relative to SSP. Although their costs also decrease over time, the interaction terms indicate that RANDOM ($\beta = -0.00340$, $p < .001$) and HSP ($\beta = -0.00382$, $p < .001$) experience slightly steeper declines. **Nevertheless, SSP maintained the lowest overall detection cost across the simulation window, consistently outperforming RANDOM and HSP by 3.16 and 4.90 cost units respectively, based on model estimates of baseline differences and subsequent trends.**

In addition to differences in cost efficiency, the policing strategies also varied substantially in their ability to accumulate serious offender detections over time. Figure 4 presents the cumulative number of serious offenders detected across the 500-step simulation for each strategy. The RANDOM strategy consistently recorded the highest cumulative detections, followed by SSP, while HSP lagged significantly with a much slower growth trajectory. Although SSP started with a lower detection rate compared to RANDOM and HSP, its detection curve steepened over time, indicating accelerating performance as behavioural patterns accumulated.

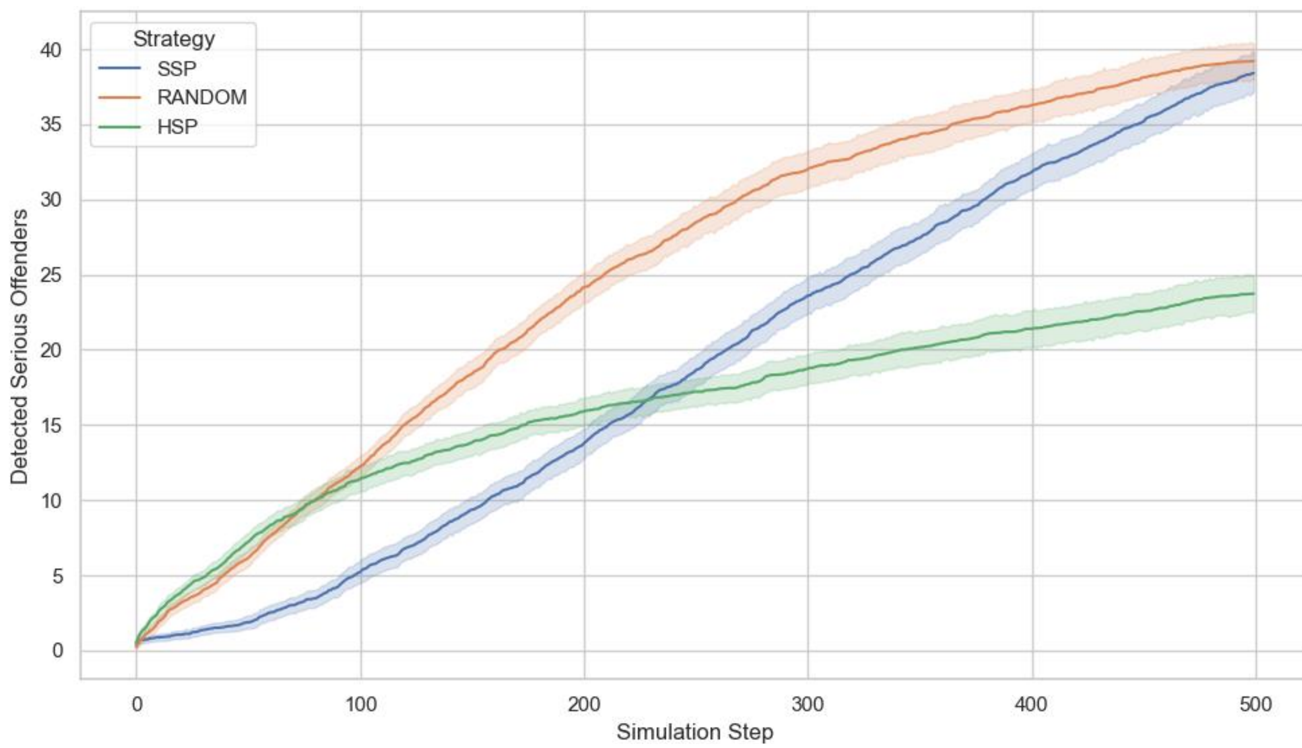

Figure 4. Cumulative number of serious offenders detected over time across police patrol strategies over 500 simulation steps, showing mean trends and variability bands.

To formally assess these dynamics, a linear regression model was fitted. The results show that although both HSP and RANDOM began with significantly higher initial detection counts relative to SSP ($\beta = 8.89$ and 7.63, respectively; $p < 0.001$), their growth trajectories diverged markedly. The interaction term for HSP × Step was strongly negative and significant ($\beta = -0.0456$, $p < 0.001$), indicating a substantially slower increase in detections compared to SSP. RANDOM also exhibited a significantly lower growth rate ($\beta = -0.00479$, $p < 0.001$), though the effect was less pronounced than for HSP. **Based on model estimates, SSP not only started with lower detection counts but also achieved a steeper growth trajectory over time, outperforming HSP and RANDOM by 0.046 and 0.0048 units per step respectively in the rate of serious offender detections.**

4.3 ANOVA Tests for Patrol Strategy Effects

The simulation results demonstrate significant differences in policing performance across the three patrol strategies, as summarized in Table 2. One-way ANOVA tests confirmed statistically significant variation across all primary metrics ($p < 0.01$), indicating that patrol strategy had a measurable impact on model outcomes.

Table 2. Comparative performance metrics across police patrol strategies (SSP, RANDOM, HSP), reporting means, standard deviations, and ANOVA F-statistics.

	SSP		RANDOM		HSP		F-statistic
Variables	M	(SD)	M	(SD)	M	(SD)	
Avg_Cost_per_Detection	**2.23***	0.22	4.30	0.55	5.61	1.20	240.78***
Detection_Precision	**0.56***	0.05	0.41	0.04	0.30	0.05	361.79***
Avg_Risk	**0.14***	0.03	0.10	0.02	0.10	0.01	62.85***
Avg_Escapes	**0.45***	0.16	1.19	1.82	0.53	0.26	7.31***
Avg_Detection_Delay	188.03	16.84	105.76	13.95	**73.73***	13.24	798.10***
Avg_Offences_per_Detected	8.38	0.67	5.44	0.56	**4.17***	0.49	692.82***
Police_Efficiency_Serious	1.92	0.25	**1.96***	0.22	1.19	0.23	174.80***
Police_Efficiency_Minor	1.53	0.19	**2.88***	0.23	2.77	0.28	510.60***
Total_Investigations	279.82	29.73	398.72	39.21	**502.02***	73.86	235.47***
Detected_Serious	38.40	4.97	**39.20**	4.46	23.74	4.52	174.80***
Active_Criminals	31.04	5.76	**3.24***	3.05	20.90	5.57	403.70***
F1-Score	0.35	N/A	**0.41***	N/A	0.30	N/A	N/A
Precision	**0.37***	N/A	0.30	N/A	0.23	N/A	N/A
Recall	0.31	N/A	**0.64***	N/A	0.43	N/A	N/A

Abbreviation: M = mean; SD = standard deviation. ∗∗∗p < .001.

In terms of cost efficiency, SSP achieved the lowest average cost per detection (M = 2.23, SD = 0.22), substantially outperforming RANDOM (M = 4.30, SD = 0.55) and HSP (M = 5.61, SD = 1.20) (F = 240.78, $p < 0.01$), confirming its advantage in resource utilization.

SSP also outperformed in detection precision, averaging 0.56 compared to 0.41 (RANDOM) and 0.30 (HSP) (F = 361.79, $p < 0.01$). Perceived offender risk was highest under SSP (M = 0.14), slightly above RANDOM and HSP (both M = 0.10) (F = 62.85, $p < 0.01$). Escape rates followed a similar pattern: SSP exhibited the lowest average (M = 0.45) compared to RANDOM (M = 1.19) and HSP (M = 0.53) (F = 7.31, $p < 0.01$).

Temporal responsiveness was also assessed. Although Avg_Detection_Delay and Avg_Offences_per_Detected showed high collinearity ($R^2 > 0.90$), both were retained to capture distinct aspects of investigative dynamics: temporal lag and cumulative offending burden. SSP recorded the longest average detection delay (M = 188.03), versus RANDOM (M = 105.76) and HSP (M = 73.73) ($F = 798.10$, $p < 0.01$), reflecting the behavioural accumulation logic. Similarly, SSP exhibited the highest average offences per detected criminal (M = 8.38), compared to RANDOM (M = 5.44) and HSP (M = 4.17) ($F = 692.82$, $p < 0.01$).

Regarding police efficiency, RANDOM achieved the highest serious crime detection per officer (M = 1.96), closely followed by SSP (M = 1.92), and HSP (M = 1.19) ($F = 174.80$, $p < 0.01$). For minor offences, RANDOM again led (M = 2.88), ahead of SSP (M = 1.53) and HSP (M = 2.77) ($F = 510.60$, $p < 0.01$), indicating broader but less selective enforcement.

In terms of investigative workload, SSP required significantly fewer investigations per simulation (M = 279.82) compared to RANDOM (M = 398.72) and HSP (M = 502.02) ($F = 235.47$, $p < 0.01$), while maintaining comparable serious offender detection (SSP: M = 38.40; RANDOM: M = 39.20; HSP: M = 23.74) ($F = 174.80$, $p < 0.01$).

The number of undetected active criminals was highest under SSP (M = 31.04), followed by HSP (M = 20.90) and RANDOM (M = 3.24) ($F = 403.70$, $p < 0.01$), highlighting SSP's focus on selectivity over comprehensive coverage.

Finally, composite performance metrics revealed key trade-offs. SSP achieved the highest precision (0.37), whereas RANDOM achieved the highest recall (0.64), outperforming SSP (0.31) and HSP (0.43). F1-scores showed RANDOM leading (0.41), followed by SSP (0.35) and

HSP (0.30), indicating that RANDOM optimized detection breadth, while SSP prioritized targeted resource deployment.

Taken together, these results highlight a fundamental trade-off between coverage and selectivity: while RANDOM maximized overall detection volume and recall, SSP achieved comparable serious offender identification with 30% fewer investigations, higher precision, and substantially lower cost per detection.

4.4 Sensitivity Analysis of SSP

4.4.1 Paired-Sample t-tests

To examine the robustness of SSP under varying operational intensities, I conducted a sensitivity analysis by defining two implementation variants: SSP_Low and SSP_High. While SSP was not designed to maximize overall detection volume, the number of serious offender identifications serves as a relevant benchmark for evaluating operational behaviour across configurations. Table 3 presents the paired-sample t-test results comparing serious offender detections across SSP strategy configurations. Under low-intensity conditions, SSP_Low detected significantly fewer serious offenders than RANDOM patrol, with a mean difference of 9.34 (SD = 6.89), representing a 22.8% reduction ($t(49) = -9.58$, $p < .001$). Compared to HSP, however, this pattern reversed: SSP_Low produced 6.12 more detections (SD = 6.12), a 29.4% increase ($t(49) = 7.07$, $p < .001$). **These results suggest that while SSP_Low detected 22.8% fewer serious offenders than RANDOM, it outperformed HSP by 29.4% under identical simulation settings, indicating its relative effectiveness compared to traditional spatial targeting at lower deployment levels.**

Table 3: Sensitivity Analysis of Detected Serious Crimes under Different SSP Strategy Configurations

Metric	SSP_Low_Inensive vs RANDOM	SSP_Low_Intensive vs HSP	SSP_High_Intensive vs RANDOM	SSP_High_Intensive vs HSP
% Change	-22.8%	+29.4%	+8.2%	+84.2%
M	-9.34	6.12	2.8	18.26
SD	6.89	6.12	5.51	6.53
95% CI	[-11.30, -7.38]	[4.38, 7.86]	[1.23, 4.37]	[16.40, 20.12]
t-value	-9.58***	7.07***	3.6***	19.77***

Abbreviation: CI = confidence interval; M = mean; SD = standard deviation. ***p < .001.

When SSP is implemented at high operational intensity, detection counts increase consistently. SSP_High detected 2.80 more serious offenders than RANDOM (SD = 5.51), an 8.2% increase ($t(49) = 3.60$, $p < .01$), and 18.26 more than HSP (SD = 6.53), an 84.2% increase ($t(49) = 19.77$, $p < .001$). These comparisons highlight the substantial gains associated with intensifying SSP's application, particularly relative to HSP.

Figure 5 further illustrates this pattern over time. The cumulative detection trajectory of SSP_High closely tracks that of RANDOM and converges around step 400, suggesting comparable levels of serious offender detection by the end of the simulation. In contrast, SSP_Low remains substantially lower throughout the simulation window, while HSP shows an early rise followed by a marked slowdown after step 200.

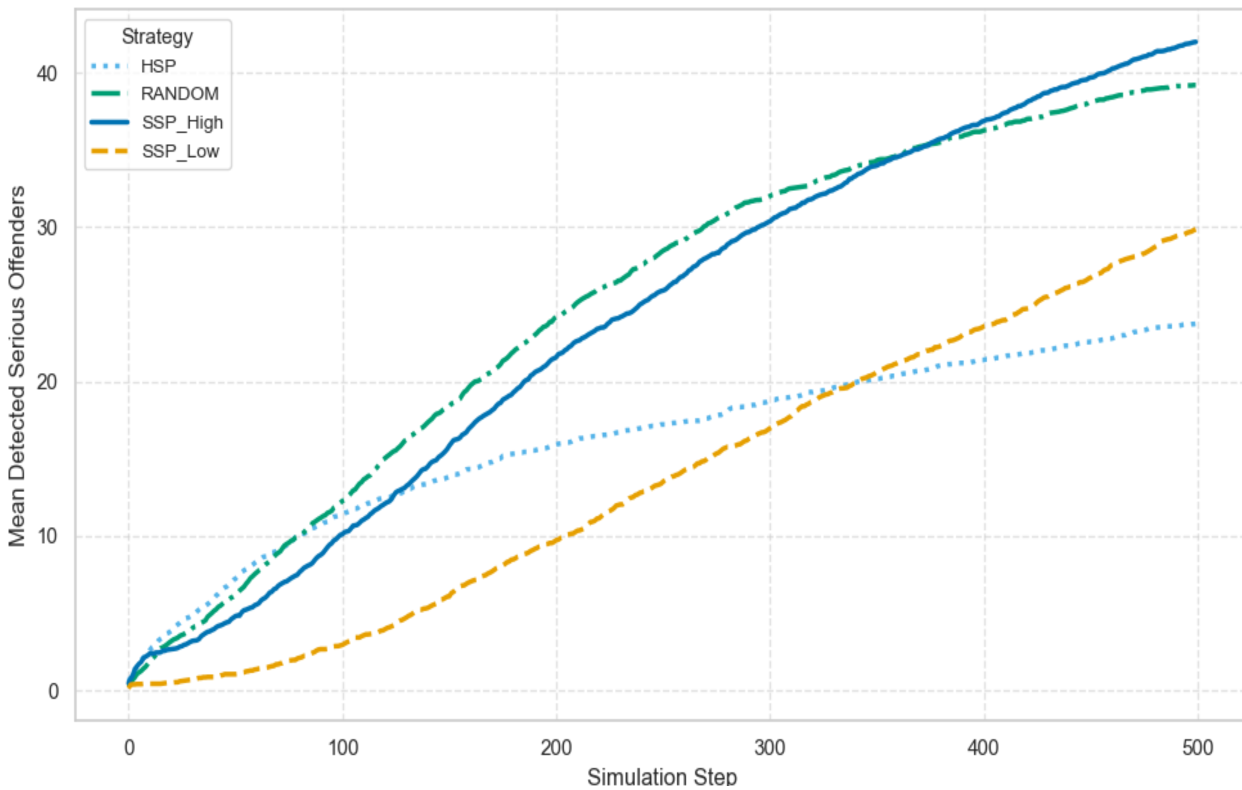

Figure 5. Cumulative number of serious offender detections across different policing strategies during sensitivity testing over 500 simulation steps.

Additionally, Table 4 reports results for total detected offenders, encompassing both serious and minor crimes. Under the low-intensity condition, SSP_Low detected 44.72 fewer individuals than RANDOM (SD = 5.68), representing a 46.2% reduction ($t = -55.65$, $p < .001$), and 27.06 fewer than HSP (SD = 7.53), a 33.9% reduction ($t = -25.43$, $p < .001$).

Table 4: Sensitivity Analysis of Detected Criminals under Different SSP Strategy Configurations

Metric	SSP_Low_Inensive vs RANDOM	SSP_Low_Intensive vs HSP	SSP_High_Intensive vs RANDOM	SSP_High_Intensive vs HSP
% Change	-46.2%	-33.9%	-11.8%	+8.2%
M	-44.72	-27.06	-11.54	6.12
SD	5.68	7.53	5.97	6.86
95% CI	[-46.33, -43.11]	[-29.20, -24.92]	[-13.24, -9.84]	[4.17, 8.07]
t-value	-55.65***	-25.43***	-13.68***	6.3***

Abbreviation: CI = confidence interval; M = mean; SD = standard deviation. ***p < .001.

Under high-intensity deployment, SSP_High detected 11.54 fewer individuals than RANDOM ($SD = 5.97$), an 11.8% difference ($t = -13.68$, $p < .001$), but outperformed HSP by 6.12 detections ($SD = 6.86$), an 8.2% increase ($t = 6.30$, $p < .001$). **From low to high intensity configurations, SSP's relative detection performance improved substantially: the total detection gap with RANDOM decreased from 46.2% to 11.8%, while its position relative to HSP shifted from 33.9% below to 8.2% above, based on model estimates.**

The dynamic trends depicted in Figure 6 further support this pattern. While RANDOM consistently maintains the highest cumulative detection curve throughout the simulation, SSP_High displays a steady upward trajectory but does not converge with RANDOM. SSP_High overtakes HSP by a substantial margin early in the simulation and maintains this advantage through step 500. In contrast, SSP_Low accumulates detections at a slower and more linear rate, remaining significantly below both benchmark strategies across the entire simulation period.

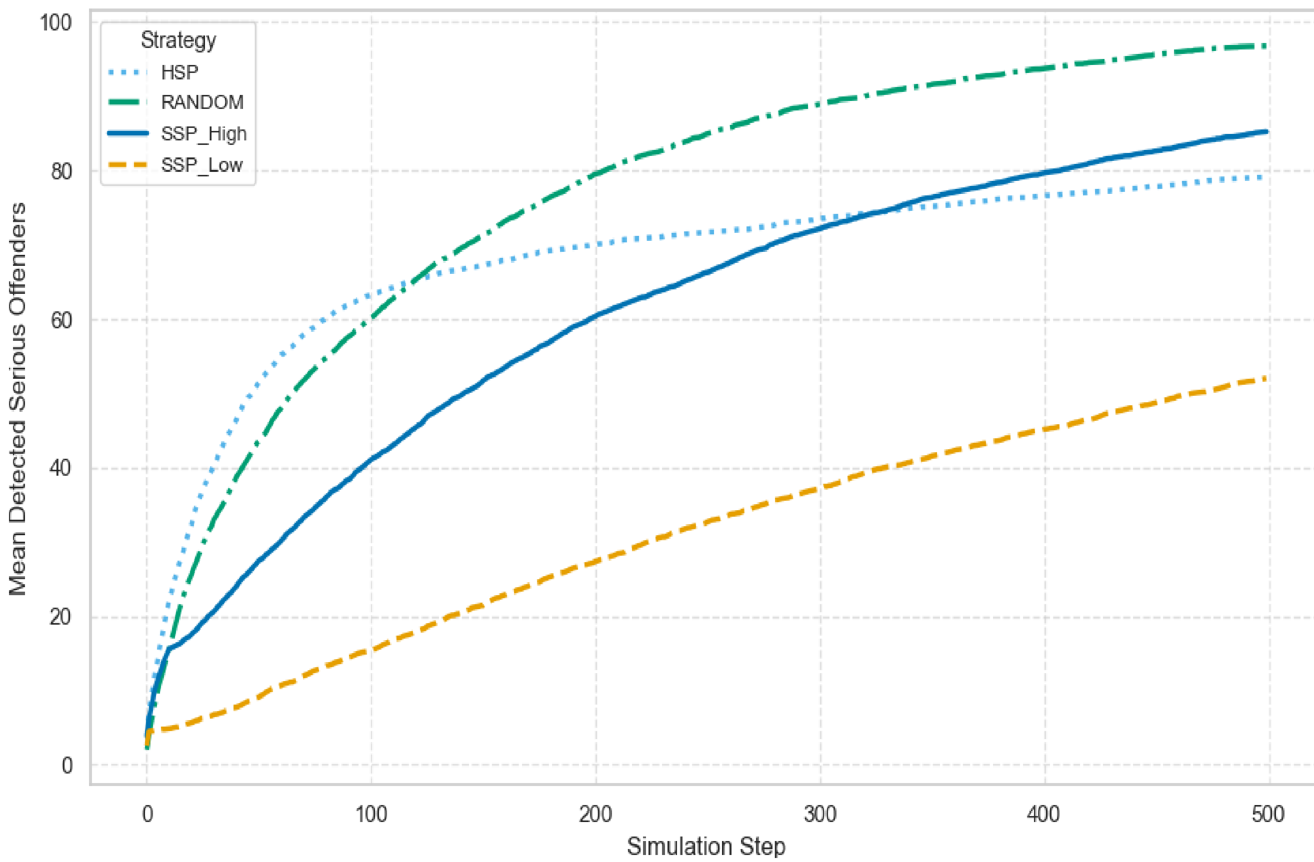

Figure 6. Cumulative number of detection of serious and minor offenders across different policing strategies during sensitivity testing over 500 simulation steps.

Taken together, the sensitivity analysis suggests that the relationship between SSP intensity and detection outcomes depends critically on the enforcement configuration. Performance differences between SSP and the benchmark strategies vary depending on whether serious or total offender detections are considered, and whether SSP_Low or SSP_High is implemented. These distinctions highlight how operational parameters shape the observed effectiveness of SSP in practice.

4.4.2 SSP Mechanism Attribution

In addition to comparing total detection outcomes across strategies, I further examined the proportion of serious offenders whose identification could be directly attributed to the internal SSP mechanism. This mechanism operates by initiating investigations based on accumulated offence histories, with agents selectively prioritising offenders according to investigation bias settings ("ssp_bias_for_minor": minor or serious offence targeting, i.e., SSP_High = 0.6, SSP = 0.8, SSP_Low = 0.9). To assess the extent to which detection outcomes were rooted in SSP's behavioural logic, as opposed to incidental or random investigative actions. I calculated the proportion of serious offender detections linked to the SSP mechanism out of the total serious offenders detected as shown in Table 5.

Table 5. Proportion of Serious Offender Detections Attributable to the Internal SSP Mechanism Across Strategy Configurations

Strategy	Mean	SD	95% CI Low	95% CI High
SSP	0.83	0.06	0.81	0.85
SSP_High	0.57	0.07	0.55	0.59
SSP_Low	**0.93***	0.05	0.92	0.94

Abbreviation: CI = confidence interval; M = mean; SD = standard deviation. ∗The highest.

Under the SSP_Low configuration, 92.9% of serious offender detections were attributed to the internal SSP mechanism (SD = 0.049; 95% CI = [0.915, 0.942]), representing the highest attribution rate. The standard SSP setting produced a mean attribution rate of 82.9% (SD = 0.059; 95% CI = [0.812, 0.846]). In contrast, the SSP_High configuration exhibited a substantially lower attribution rate of 57.2% (SD = 0.073; 95% CI = [0.552, 0.593]). **SSP_Low achieved an attribution rate 35.7 percentage points higher than SSP_High, highlighting how parameter settings critically shape the alignment between detection outcomes and SSP's intended logic.**

These findings reveal considerable variation in the proportion of mechanism-driven detections depending on operational intensity. In particular, the contrast between the high attribution rate under SSP_Low and the significantly lower rate under SSP_High highlights the influence of configuration parameters on how closely detection outcomes align with the intended behavioural logic of SSP. The implications of these patterns are explored further in the following discussion.

5. Discussion

5.1 Overall Performance of SSP Compared to Other Strategies

Across both experiments, Self-Selection Policing (SSP) demonstrated a markedly different detection profile, prioritising quality over quantity compared to Random Patrol and Hot Spots Policing (HSP). Although SSP was not intended to maximise detection volume, it significantly improved detection quality, measured by the proportion of serious offenders identified. The high-intensity variant outperformed Random by 8.2% and HSP by 84.2% in this regard. Notably, these improvements were accompanied by the lowest average cost per detection (M = 2.23) and the highest detection precision (0.56), outperforming Random (0.41) and HSP (0.30).

In addition, SSP required fewer investigations overall (M = 279.82), compared to Random (M = 398.72) and HSP (M = 502.02), indicating a higher level of investigative efficiency. However, these advantages came at the cost of comprehensiveness. SSP exhibited the lowest recall (0.31), meaning that although it accurately confirmed serious offenders when sufficient evidence was present, it failed to identify the majority of such individuals.

These findings highlight a critical trade-off in SSP's recall performance. Although the model achieves high precision, accurately identifying high-risk individuals once sufficient behavioural evidence accumulates, its recall remains strikingly low (0.31). As a result, a large proportion of serious offenders remain undetected. This is reflected in the simulation: under SSP, the number of active serious criminals remains high (M = 31.04), compared to only 3.24 under Random Patrol. This discrepancy is not due to reduced offender risk, but rather because their behavioural patterns do not accumulate quickly or visibly enough to trigger investigative attention, largely due to SSP's longer investigation delays compared to Random and HSP. These findings are consistent with the view that crime and criminality often resemble an iceberg structure, where most of the threat lies beneath the surface. Many high-risk individuals may never commit visible infractions, or do so infrequently enough to avoid detection (Rossmo, 2020; Roach and Pease, 2016).

5.2 Why SSP Works and Where It Fails

This result highlights another important point: high precision reflects a higher proportion of serious crimes among detected cases compared to the other two simulated strategies. However, a key challenge to the effective implementation of SSP, as discussed in the literature, is the risk of false positives (Roach and Pease, 2016; Rossmo, 2020; Farquharson, 2020). In principle, trigger offences could direct police attention towards active offenders and serious criminal activity. However, in practice, many individuals who are not of law enforcement interest also commit these offences. This is partly because non-offenders significantly outnumber offenders, especially when minor infractions are excluded (Rossmo, 2020).

These findings reveal a paradox: SSP achieves low false positives and high precision precisely because it intervenes only when behavioural signals accumulate beyond a threshold. While this reflects efficient investigative logic, it imposes clear limitations from a deterrence perspective. As Braga and Weisburd (2011) argue, deterrent effects depend on the perceived risk of sanction, not merely on the actual occurrence of enforcement. In my ABM, offenders make decisions based on situational opportunities and their perceived risk of police presence, consistent with rational choice models of offending (Brantingham and Brantingham, 2013; Clarke and Cornish, 2016; Gibbs, 1975; Zimring and Hawkins, 1973). Under SSP deployment, most offenders in the simulation remain active after 500 steps and show an increased tendency to commit serious offences, largely due to longer investigation delays.

However, because SSP only investigates individuals highly likely to have committed serious crimes, most agents perceive little or no enforcement presence in their environment (Braga and Weisburd, 2011). Consequently, while SSP avoids unnecessary interventions, it also fails to propagate perceived sanction risk broadly, thereby weakening its general deterrence effect. In this way, the very feature that ensures high investigative precision—namely, low false positives—ultimately undermines the broader behavioural impact of the strategy.

SSP's low recall thus reveals a deeper structural issue: the model excels at identifying those already within surveillance networks while systematically overlooking individuals outside learned behavioural patterns. This observation aligns with findings from predictive policing research, which show that such systems predict the subset of crime already visible to police, not crime itself (Lum and Isaac, 2016). The result is a feedback loop where serious crime detection is high, but the detection band remains narrow. Precision is achieved at the cost of comprehensiveness and early intervention (Lum and Isaac, 2016). SSP risks reinforcing visibility bias, detecting those who appear detectable rather than those most in need of intervention.

Moreover, the high precision observed in the SSP model, particularly in its high-intensity configuration, may initially suggest strong performance. However, such precision often results from narrow algorithmic targeting that emphasises observable correlations rather than underlying causal structures (Meijer and Wessels, 2019). In this simulation, SSP's efficiency in detecting prolific offenders at low investigative cost relies on repeated behavioural patterns, but excludes individuals whose risk signals do not conform to learned archetypes.

Furthermore, Meijer and Wessels (2019) caution that reliance on data-driven systems can obscure accountability, as officers may defer to algorithmic judgements without critically engaging with underlying assumptions. In the context of SSP, this raises operational concerns: precision may be achieved without sufficient reflection on whom the model fails to detect. Rather than indicating efficacy, high precision in predictive models may signal an overconfident narrowing of the investigative lens (Perry, 2013; Meijer and Wessels, 2019).

Ultimately, the trade-off between precision and recall in SSP is not inherently problematic but reflects a question of strategic fit. Where detection quality and investigative efficiency are prioritised, SSP offers a compelling model. However, its narrow scope limits its preventive capacity. SSP should thus be understood not as a comprehensive crime control strategy, but as a targeted mechanism within a broader policing toolkit. The key challenge is to configure SSP in a way that maintains its precision while mitigating its blind spots—a question further explored through the comparison between SSP_High and SSP_Low configurations.

5.3 Sensitivity Testing: Effects of Different SSP Intensities

Under the "low-intensity" SSP configuration, key parameters such as investigation probability, detection success rate, and memory window are conservatively set. The model therefore operates primarily along its intended behavioural pathway, with investigations triggered only when closely spaced minor offences accumulate. Detection outcomes are largely attributable to the internal mechanism, resulting in high mechanism fidelity. However, because parameter strengths are weak, the mechanism is triggered infrequently, allowing many high-risk offenders to fall outside its reach. Consequently, random variation and uncontrolled behaviours dominate, limiting overall effectiveness. Although the mechanism remains “pure”, supported by a relatively high ssp_bias_for_minor value that directs police attention towards minor offences, its operational coverage is too narrow to generate substantial macro-level impact.

In contrast, the “high-intensity” SSP configuration employs more aggressive parameter settings, including a higher probability of investigation and an extended memory window. This significantly increases the frequency of interventions, resulting in a visible reduction in serious crime and achieving high strategic effectiveness. However, not all reductions can be directly attributed to the original behavioural logic.

Strong interventions are likely to activate multiple operative pathways. The intended mechanism, which involves flagging and escalating repeat minor offenders, remains functional. At the same time, broader enforcement efforts generate generalised deterrence and incidental apprehensions (Greenwood-Lee et al., 2016). As shown in complex systems models of public health interventions, even policies designed with a clear logic may trigger unforeseen adaptations in dynamic environments, ultimately diverging from their intended pathways (Greenwood-Lee et al., 2016). Agents may respond strategically to intense enforcement, adjusting their behaviours to optimise outcomes. Although the results may appear promising, they often bypass the mechanisms originally designed to activate SSP.

In this simulation, some crime prevention may result from offenders' heightened perceived risk or from random officer interventions, rather than from the specific “minor-to-serious” escalation logic. As the theory of multiple realisability suggests, high-intensity interventions often produce macro-level outcomes through the convergence of various micro-level processes (Sawyer, 2005). Thus, strong interventions tend to operate through both intended and unintended pathways, weakening the exclusive explanatory power of any single mechanism. Accordingly, compared to SSP_Low, the mechanism dominance of SSP_High becomes diluted. Although the high-intensity configuration is more effective overall, a smaller proportion of outcomes can be traced directly to the internal behavioural mechanism, with a growing share driven by random enforcement effects.

5.4 Limitations and Implications for Future Research

In future implementation, this research suggests that SSP’s selective focus may be more appropriate in intelligence-led operations or in units targeting serious crime, where the objective is not to broadly suppress crime but to surgically disrupt high-harm individuals. In such contexts, high precision is not a limitation but a design imperative, allowing law enforcement to concentrate efforts on targets with the highest potential impact. As Kennedy (2012) observes in his analysis of Operation Ceasefire, some of the most effective deterrent actions did not involve the harshest penalties but rather minor steps strategically deployed against groups with elevated violence potential. The impact stemmed not from severity but from focused visibility and credibility. What mattered was who was targeted and how consistently, not how many were swept in (Kennedy, 2012).

This approach reflects the strategic thinking of Durlauf and Nagin (2011), who argue that when enforcement resources are limited, focusing on the certainty of being caught rather than the severity of punishment yields stronger deterrent effects. SSP aligns with this logic. Rather than spreading resources thin across the entire population, it concentrates attention on individuals who cross defined behavioural thresholds, reducing unnecessary interventions while still identifying those most likely to cause serious harm (Kennedy, 2012). The result is a more cost-effective use of policing effort, as shown by SSP's lowest average cost per detection and a higher perceived risk (M = 0.14), compared to Random Patrol and HSP, both at M = 0.10. By making enforcement more predictable and targeted, SSP does not need to cast a wide net to influence behaviour. Its narrow focus is not a flaw; it is a practical way to achieve greater impact with fewer actions (Durlauf and Nagin, 2011).

The ABM captures how offenders assess local enforcement presence and adjust their offending decisions based on perceived sanction risk. However, this deterrence mechanism may extend beyond individuals to co-offending groups. When one group member is sanctioned, the increased scrutiny on the entire group can create collective accountability, shifting group norms and incentivising members to discourage further violence (Tillyer and Tillyer, 2019; Tillyer and Kennedy, 2008; Braga and Weisburd, 2012). This suggests that perceived risk is shaped not only by personal experience but also by social proximity to enforcement actions within the network. This effect is well captured by group-based deterrence dynamics in focused deterrence strategies.

Therefore, in ABM frameworks, modelling deterrence at the group level rather than solely through individual histories may offer a more realistic representation of how targeted enforcement ripples through offender networks. Integrating network-level feedback mechanisms could enhance understanding of how policing approaches like SSP exert influence beyond directly sanctioned agents, potentially amplifying deterrent effects through informal group dynamics.

It is important to note that the assumptions embedded in my simulation are based on an abstracted model of urban policing and should be interpreted with caution (Gilbert and Troitzsch, 2005; Groff, 2007a, 2007b; Weisburd et al., 2017). My agent-based model was not designed to replicate SSP's literal operational form, in which trigger offences are often used to justify deeper investigation. Instead, it was intended to examine the underlying strategic logic of SSP—namely, the prioritisation of high-risk individuals based on behavioural accumulation. Rather than relying on static offence categories such as traffic violations to trigger a downstream detection sequence (Roach and Pease, 2016), the simulation implements a deterrence-based decision model. Offenders assess their immediate environment and the perceived sanction risk before deciding whether to engage in serious offending. In this sense, what is tested is not the institutional mechanics of SSP per se, but the principle of behavioural filtering under varying thresholds of intervention intensity.

The grid-based environment in the model represents a simplified spatial abstraction, composed of a city with 100 by 100 cells. Each cell aggregates localised social activity, criminal decision-making, and policing encounters. While this design allows the model to represent geographic dispersion and clustering of risk, it does not capture the topological complexity of actual urban infrastructure, such as road networks, administrative boundaries, or residential zoning. The model also does not incorporate political or legal constraints that would influence how SSP is deployed in practice. For this reason, I do not make direct claims about the feasibility of SSP at the city scale. Whether a mechanism like SSP can generate sustained impact across multiple jurisdictions remains an open question, especially in light of potential concerns about public legitimacy and resource constraints that may accompany its expansion.

Nevertheless, the use of ABM in this context allows for systematic exploration of the latent potential of SSP-type logic under controlled and replicable conditions. ABM enables the examination of how variation in key parameters, such as investigation bias, memory window, or threshold sensitivity, affects detection outcomes, detection cost, and strategy precision. The model also reveals trade-offs that may not be visible in descriptive field data. For instance, increasing investigative intensity raises overall detection rates but reduces mechanism purity (defined here as the proportion of detections directly attributable to SSP's designed behavioural mechanism). Conversely, reducing bias towards minor offences may improve theoretical consistency, yet fails to generate practical returns. These tensions suggest that SSP, like many strategic policing approaches, does not operate as a fixed template but as a parameter-sensitive framework (Weisburd et al., 2017; Weisburd et al., 2010; Malleson et al., 2010). Its effectiveness depends on careful alignment between theoretical logic and the conditions under which it is deployed.

In the end, I cannot assume that the strategic advantages of SSP as observed in simulation will naturally translate into effective or legitimate practice across real-world jurisdictions. Carefully designed field studies are necessary to test these questions in applied settings. This would require jurisdiction-level experiments or, alternatively, comparative trials in which some enforcement units adopt SSP logic while others maintain conventional practices. This study demonstrates that ABM can support the design and justification of field studies, rather than replace them. ABM can help identify the most effective level of intervention intensity to test and provide insight into how patrol resources should be allocated across time and space (Weisburd et al., 2017). Field experiments are often difficult to develop, as the appropriate trigger offences remain under exploration (Roach and Pease, 2016; Farquharson, 2020). This research suggests that ABM can play a valuable role in prototyping such studies by refining hypotheses, clarifying assumptions, and improving the efficiency and realism of intervention design.

6. Conclusion

The agent-based model developed in this study enabled a counterfactual evaluation of Self-Selection Policing (SSP) within a dynamic offender–policing environment. Findings indicated that SSP strategies increased the precision of serious offender detection and reduced investigative costs compared to random and hot spots patrols, albeit with a trade-off in overall coverage. Building on theoretical foundations emphasising offender versatility and situational crime patterns (Roach and Pease, 2016; Cornish and Clarke, 2008; Cohen and Felson, 1979; Brantingham and Brantingham, 1993), and supported by simulation-based analysis (Eck and Liu, 2008; Groff and Birks, 2008), this study provides preliminary mechanism-level evidence for SSP as a targeted investigative strategy. However, limitations arising from behavioural simplifications and static environmental structures warrant caution in extrapolating the results to real-world settings. Overall, the findings suggest that the behavioural logic underlying SSP merits further empirical investigation through both refined simulations and applied field studies.

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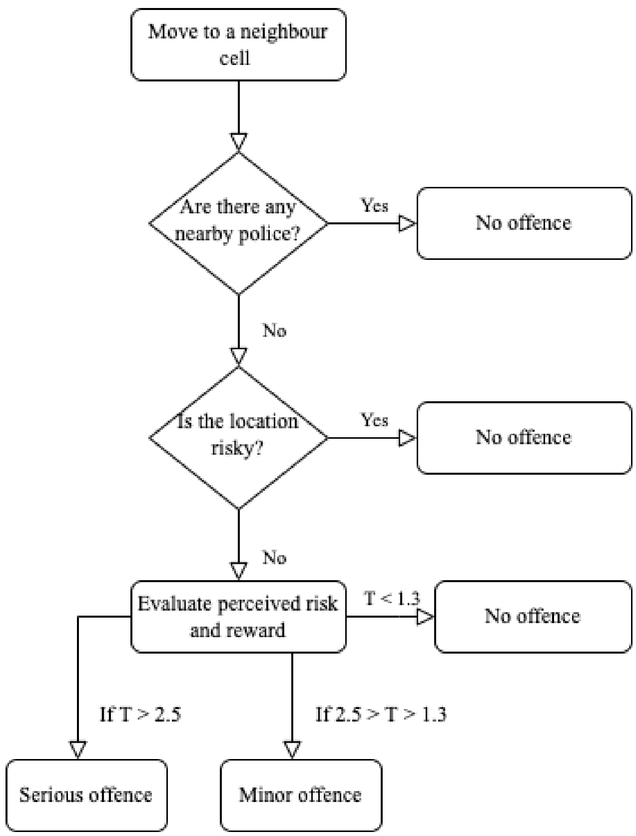
Appendices

Appendix 1. Agent Decisions in the Model

First, the agent moves to a randomly selected neighbouring grid cell. This mobility step reflects routine activity perspective (Cohen and Felson, 1979), in which spatial movement increases the likelihood of encountering opportunities and guardianship (Weisburd et al., 2017). Upon arriving at a new cell, the agent scans the surrounding area for police presence. If any police are found within a specified radius, no offence occurs. The presence of police is interpreted as a direct deterrent, operationalizing the "capable guardian" element of the routine activity framework.



Note: This table is a duplicate of Figure 1 in the main text, included here for completeness.

Second, if no nearby officers are detected, the agent evaluates whether the location is inherently risky. Grid cells characterized by recent enforcement or elevated deterrent cues are classified as "risky," and agents encountering these zones will abstain from offending. This additional check represents an indirect risk filter and captures how agents internalize environmental signals over time.

Third, the offence score integrates factors related to the offence type (reward), environmental opportunity, and the agent's individual perception of policing risk (Weinsburd et al., 2017). A criminal action is initiated only when the evaluated score surpasses a set threshold for action. The offence threshold T_i for offence type where i ∈ {serious, minor}. is defined as:

$$Threshold = \underbrace{Reward}_{\text{offence type}} \cdot \underbrace{Opportunity(x, y, t)}_{\text{grid cell}} \cdot \underbrace{Propensity \cdot PercievedRisk}_{\text{individual} \cdot \text{environment}} \cdot \alpha$$

$$T_i = R_i \cdot O(x, y, t) - \lambda \cdot PR \cdot \alpha_i$$

R_i is the reward associated with the offence type (3.0 for serious offences and 1.0 for minor offences), while O (x, y, t) represents the opportunity factor of the grid cell at location (x, y) and time t, which ranges from 0.5 to 1.5. The product λ · PR · α_i represents the total perceived penalty, where λ is the agent-specific propensity parameter indicating the initial motivation to commit crimes, uniformly sampled from (0.5, 1.5) at model initialization, PR is the dynamically updated perceived policing risk, and α_i is the punishment multiplier (3.0 for serious and 0.5 for minor offences, respectively).

A criminal agent only proceeds to offend if the computed score meets the decision logic illustrated below. The decision framework ensures that serious offences occur only when they are substantially more rewarding than minor alternatives, and when the agent's probabilistic disposition toward serious offending is met:

$$Decision = \begin{cases} Serious\ Offence, & if\ T_{serious} - T_{minor} > 2.5 \text{ and } rand(0,1) < P_{serious} \\ Minor\ Offence, & if\ T_{minor} > 1.3 \\ No\ Offence, & Otherwise \end{cases}$$

The probability threshold P_serious represents the agent's inherent propensity toward committing serious offences, which is independently initialized for each agent from a uniform distribution ranging between 0.3 and 0.7. Even if the serious offence score exceeds the minor offence score by more than 2.5, the action is only executed if a uniformly generated random value in (0, 1) falls below this threshold. This stochastic gate introduces behavioural heterogeneity and reflects the idea that not all offenders with access to high-gain opportunities will necessarily pursue the most aggressive path.

Appendix 2. Model Parameters

The table below lists all key parameters used in the simulation model, with descriptions and corresponding values or value ranges extracted directly from the source code. These parameters govern agent behaviour, patrol strategy, and environmental settings.

Parameter	Description (including value or range)
cover_prob	Probability of avoiding detection after being investigated (random.uniform(0.2, 0.8))
offending_propensity	Internal variable adjusted based on past detection and offending history (dynamic)
activity_level	Probability of attempting offence each step (random.uniform(0.4, 0.9))
serious_prob	Probability of committing a serious offence when utility threshold is met (random.uniform(0.3, 0.7))
minor_reward	Fixed reward for committing a minor offence (1.0)
serious_reward	Fixed reward for committing a serious offence (3.0)
risk_sensitivity	Agent's individual sensitivity to perceived policing risk (random.uniform(0.5, 1.5))
perceived_risk	Agent's dynamically updated perception of police risk based on nearby officers (updated per step)
cooldown_steps	Cooldown period after offending: 10 steps (minor), 20 steps (serious)
serious_offence_count	Number of serious offences previously committed by the agent (accumulative integer)
strategy_type	Patrol strategy assigned to the police agent (SSP, HSP, or RANDOM)

vision_radius	Radius (in cells) within which police detect nearby criminals (default = 10)
investigate_prob	Probability of initiating investigation in RANDOM and HSP mode (default = 0.3)
success_prob	Probability of successful detection upon investigation in RANDOM and HSP mode (default = 0.8)
ssp_bias_for_minor	Bias toward investigating minor offenders in SSP strategy (i.e., 0.9 for SSP_Low, 0.8 for SSP, 0.6 for SSP_High)
ssp_investigate_prob	Probability that SSP investigates a known offender (default = 0.5)
ssp_success_prob	Probability of successful detection under SSP (default = 0.8)
ssp_memory_window	Window (in steps) to consider recent offending history in SSP (default = 2)
opportunity_map	Spatial grid with opportunity values randomly drawn from uniform(0.5, 1.5)
hotspot_cells	Set of high-opportunity or high-risk cells used in HSP patrol logic (dynamically assigned)
COST_MINOR_INVESTIGATION	Cost assigned to investigating a minor offence (1.0)
COST_SERIOUS_INVESTIGATION	Cost assigned to investigating a serious offence (3.0)
random_seed	Random seed used to initialize simulation randomness (default = 42)
N_criminals	Number of criminal agents initialized in the simulation (typically 100)
N_police	Number of police agents initialized (typically 20)

width / height	Dimensions of the square grid environment (default = 100 × 100)

Appendix 3. SSP Model Logic

In the SSP condition, each police agent patrolled the grid following a predefined investigative bias, specified by the ssp_bias_for_minor parameter. This bias determined whether the officer prioritized identifying offenders associated with minor offences or serious offences. At each simulation step, police agents inspected their current grid cell for criminal agents whose behavioural records, stored within a time-limited memory window (ssp_memory_window), matched the prioritized offence type. If one or more matching offenders were present, a target was selected for potential investigation.

The decision to initiate an investigation was probabilistic and governed by the ssp_investigate_prob parameter. If an investigation was initiated, its success was determined by the ssp_success_prob parameter. Following a successful investigation, the police agent reviewed the offender's behavioural history within the memory window. An offender was classified as a serious crime detection if more than two serious offences had been committed during the memory period. The offence currently triggering the investigation was not included in this count to maintain independence between the triggering event and the historical behavioural assessment. Once classified as a detected serious offender, the agent was immediately removed from the simulation. Regardless of the investigation outcome, every investigative action incurred a fixed operational cost.

Based on the general observation that responses to serious crimes typically require more resources than responses to minor offences (Hunt et al., 2019), the model assigned investigation costs of 1.0 unit for minor offences and 3.0 units for serious offences. These values were introduced as modelling assumptions to reflect relative differences in enforcement efforts but were not directly empirically calibrated. Investigative costs were systematically recorded throughout the simulation to allow evaluation of the overall cost-efficiency of each policing strategy.

Appendix 4. Sensitivity Testing

In order to assess the sensitivity of self-selection policing (SSP), three levels of intensity towards active criminals were implemented as distinct experimental conditions: SSP_Low, SSP, and SSP_High. These variants differ in terms of both the decision thresholds and memory depth governing investigative behaviour.

The SSP_Low condition represents a lenient and opportunistic enforcement profile, with a relatively low probability of initiating investigations (p = 0.7), limited investigative success (p = 0.7), and a strong preference toward minor offenders (bias = 0.9). Additionally, SSP_Low agents only retain memory of recent offences for 1 step, reflecting minimal attention to historical behavior. The baseline SSP configuration increases investigative activity and accuracy, using investigate_prob = 0.9, success_prob = 0.8, and a more balanced bias (bias = 0.8) that allows moderate targeting of serious offenders. The memory window is extended to 2 steps. Finally, the SSP_High condition implements an intensive enforcement model. Officers initiate investigations with very high probability (p = 0.95), achieve higher detection rates (p = 0.9), and shift targeting toward serious offenders (bias = 0.6). Most notably, SSP_High agents retain memory of past offences for up to 10 steps, enabling broader temporal surveillance and repeat detection of chronic offenders. These three variants operationalize patrol intensity along four dimensions: initiative, accuracy, targeting bias, and temporal reach, offering a structured way to examine the sensitivity of the SSP mechanism under different real-world enforcement capacities.

Appendix 5. Model Output

Variable	Description
Strategy	The patrol strategy used in the simulation (\`SSP_Low\`, \`SSP_High\`, \`SSP\`, \`HSP\`, or \`RANDOM\`).
Seed	Random seed used for this simulation run to ensure reproducibility.
Step	The current simulation step (time tick).
Total_Investigations	Total number of investigations initiated by police agents. Incremented by 1 whenever

	a police agent attempts an investigation, regardless of success.
Detection_Precision	Ratio of serious offenders among all detected criminals. Calculated as: \`detected_serious / detected_criminals\` (0 if no criminals detected).
Avg_Detection_Delay	Average number of steps between a criminal's first offence and their detection. Calculated over all detected criminals: \`sum(detected_step - first_offence_step) / detected_criminals\`.
Avg_Offences_per_Detected	Average number of offences committed before detection among detected criminals. Calculated as: \`sum(number of offences for detected criminals) / detected_criminals\`.
Avg_Risk	Average perceived risk (\`perceived_risk\`) across all active criminals. Calculated as: \`sum(perceived_risk of active criminals) / number of active criminals\`.
Avg_Escapes	Average number of escapes (failed investigations) per active criminal. Calculated as: \`sum(escape_count of active criminals) / number of active criminals\`.
Police_Efficiency	Number of serious offenders detected per police officer. Calculated as: \`detected_serious / total_police\`.
Active_Criminals	Number of criminals who are still active (not yet detected and removed).

Detected_Serious	Total number of detected (captured) serious offenders.
Detected_Criminals	Total number of detected criminals, regardless of offence seriousness.
Avg_Cost_per_Detection	Average investigation cost per detected serious offender. Each minor investigation costs 1, serious investigation costs 3. Formula: `total_investigation_cost / detected_serious` (0 if no serious offenders detected).
False_Positives	Number of investigations that failed to capture a criminal. Specifically, incremented whenever an investigation attempt does not succeed in detecting a criminal (`model.false_positives += 1`).
Active_Serious_Remaining	Number of criminals who are both active and classified as serious offenders (serious offence count ≥ 2). Computed as: `len([a for a in criminal_agents if a.is_active and a.is_serious_offender])`.
Total_Serious_Offences	Cumulative number of serious offences committed by all criminals up to the current step.
Total_Minor_Offences	Cumulative number of minor offences committed by all criminals up to the current step.
Total_Offences	Cumulative total number of offences (serious + minor) committed so far.
Detections_per_10_Investigations	Number of serious offenders detected per 10 investigations. Formula:

	`(detected_serious / total_investigations) * 10` (0 if no investigations).
SSP_Resolved_Serious	Number of serious offenders detected under SSP strategies specifically via initial minor offence investigations. This measures SSP's effectiveness at early intervention.
Investigation_Success_Rate	Proportion of investigations that resulted in a successful capture of a criminal. Formula: `successful_investigations / total_investigations` (0 if no investigations).

Appendix 6. Full Python Code

Full Python code available at GitHub: https://github.com/Royywangg233/ssp-abm-dissertation.git